Example Garage

DOCUMENTATION SET BRIEFING

Vehicle Rebuild Workshop Manual

Technical Manual

`examples › workshop-manual-template › docs › vehicle-rebuild`

DOCUMENTATION SET BRIEFING

Generated: *14 October 2026*

Document ID: DOC-BRF-VEH-30001/26

© 2026 Example Garage

Example Garage — Executive Briefing

LEVEL: DOCUMENTATION SET

`briefkit > examples > workshop-manual-template > docs > vehicle-rebuild`

Generated by briefkit

Table of Contents

Scope

Procedures

1. Foreword
2. How To Use
3. Technical Data
4. Engine Block
5. Torque Master
6. Index

Reference Tables

© 2026 Example Garage

Scope

OEM-style workshop manual demonstrating the BriefKit `manual` template applied at scale to a technical service document. The structure mirrors the Volkswagen factory workshop manual and the Bentley Publishers service manual conventions.

Applicability

#	Subsystem	Coverage
1	Foreword	See Procedures section
2	How To Use	See Procedures section
3	Technical Data	See Procedures section
4	Engine Block	See Procedures section
5	Torque Master	See Procedures section
6	Index	See Procedures section

© 2026 Example Garage

Procedures

1. Foreword

This workshop manual is intended for the competent owner, enthusiast, and independent workshop technician undertaking a comprehensive engine and drivetrain rebuild. It demonstrates the OEM service-manual pattern for BriefKit, structured as numbered service groups with consistent per-chapter sections.

How This Manual Is Organized

Content is divided into eight major parts:

Step 1: **Part 1 — General Information** (Groups 01 to 06): foreword, usage notes, abbreviations, safety, technical data, special tools

Step 2: **Part 2 — Engine Mechanical** (Groups 10 to 19): block, head, lubrication, cooling

Step 3: **Part 3 — Engine Management** (Groups 20 to 28): fuel, turbo, injection, exhaust, ignition

Step 4: **Part 4 — Drivetrain** (Groups 30 to 40): clutch, gearbox, axles

Step 5: **Part 5 — Removal and Installation** (Groups 50 to 55): engine drop, timing belt

Step 6: **Part 6 — Configuration Options** (Groups 70 to 75): variant paths, ECU coding

Step 7: **Part 7 — Build Execution** (Groups 80 to 84): build sequence, commissioning, maintenance

Step 8: **Part 8 — Technical Appendices** (Groups 90 to 98): torque data, part numbers, index

Edition Information

Item	Detail
Edition	1.0
Publication date	Example
Format	BriefKit manual template, navy preset
License	CC0 1.0

2. How To Use

Chapter Structure

Each technical chapter in this manual follows a consistent structure modeled on factory workshop manual conventions.

Section	Purpose
Technical Data	Key dimensions, clearances, and operating parameters
Special Tools	Factory tool numbers and aftermarket equivalents

© 2026 Example Garage

Section	Purpose
Tightening Torques	Consolidated torque specification table
Removing	Disassembly procedure as numbered steps
Installing	Reassembly procedure as numbered steps
Testing and Checking	Measurements and acceptance criteria
Notes	Technical notes, safety callouts, cross references

Reading Conventions

Numbered steps in procedures are to be performed in the order given unless a note explicitly permits parallel work.

Cross references are written as → **Group NN Title** to direct the reader to related service groups.

Three callout levels appear in this manual:

Step 1: **DANGER** — risk of serious injury or death

Step 2: **WARNING** — risk of component damage or procedural failure

Step 3: **NOTE** — useful technical information

All dimensions are in metric (mm, Nm, kg, L). Imperial equivalents are given where commonly used by aftermarket suppliers.

3. Technical Data

Quick-reference summary of the most important technical data. For complete data see Group 90 in the appendices.

Engine

Item	Specification
Displacement	2008 cc (built)
Bore	83.01 mm
Stroke	92.80 mm
Compression ratio	9.3:1
Firing order	1-3-4-2
Target peak power	500 wheel horsepower
Target peak torque	520 Nm at wheels
Fuel	E85 flex
Redline	7200 rpm

© 2026 Example Garage

Fluid Capacities

Fluid	Capacity	Specification
Engine oil with filter	4.2 L	5W40 full synthetic
Coolant	7.2 L	OAT pink
Transmission	2.3 L	75W90 GL4
Brake fluid	1.0 L	DOT 4

Performance Targets

Parameter	Target
Peak boost pressure	25 psi
Peak EGT under load	< 950 C
Peak AFR on E85	11.8:1
Maximum ignition retard	0 degrees

4. Engine Block

Short block disassembly, machining, and assembly. Demonstrates the per-chapter structure: Technical Data → Special Tools → Tightening Torques → Procedures → Notes.

13.1 — Technical Data

Item	Specification
Block material	Grey cast iron
Built bore	83.01 mm ± 0.005 mm
Built stroke	92.80 mm
Built displacement	2008 cc
Main bearing radial clearance	0.018 to 0.038 mm
Rod bearing radial clearance	0.015 to 0.040 mm
Crankshaft endplay	0.070 to 0.230 mm
Rod side clearance	0.10 to 0.30 mm
Piston to wall (forged)	0.089 to 0.114 mm
Top ring end gap (turbo)	0.22 to 0.42 mm

© 2026 Example Garage

13.2 — Special Tools

Tool	Purpose
Rod bolt stretch gauge	ARP rod bolt installation
Plastigauge set	Bearing clearance check
Outside micrometer 0 to 100 mm	Crank journal measurement
Dial bore gauge	Cylinder measurement
Ring compressor 83 mm	Piston installation
Dial indicator with magnetic base	Crank endplay

13.3 — Tightening Torques

Fastener	Torque	Method	Reuse
Main bearing bolt OEM stretch	65 Nm + 90 deg	Stretch	No
Rod bolt OEM stretch	30 Nm + 90 deg	Stretch	No
Rod bolt ARP2000	Stretch 0.0063 inch	Gauge	Yes
Crank pulley bolt	90 Nm + 90 deg	Stretch	No
Flywheel to crank	60 Nm + 90 deg	Stretch	No
Oil pump to block	15 Nm	Standard	Yes
Oil pan to block	15 Nm	Center out	Yes

13.4 — Crankshaft and Main Bearings — Installing

Step 1: Wipe crank main journals with lint-free cloth and acetone.

Step 2: Install upper main bearing shells in the block saddles, aligning locating tangs.

Step 3: Lay the crankshaft into the block dry, no oil.

Step 4: Set Plastigauge across each main journal perpendicular to the crank axis.

Step 5: Install lower main bearing shells in the main caps.

Step 6: Torque main bolts to 65 Nm + 90 degrees without rotating the crank.

Step 7: Remove caps and read clearance from the Plastigauge calibration card.

Step 8: Target main bearing clearance: 0.018 to 0.038 mm.

Step 9: Remove the crankshaft and clean Plastigauge residue.

Step 10: Apply assembly lube to upper and lower bearing shells.

Step 11: Reinstall the crankshaft.

Step 12: Reinstall main caps and torque to final spec.

Step 13: Verify the crankshaft spins freely by hand.

Step 14: Measure crankshaft endplay with a dial indicator at the end of the crank.

13.5 — Connecting Rods and Pistons — Installing

Step 15: Verify all ring gaps are within specified range with the ring squarely in the bore.

© 2026 Example Garage

Step 16: Install rings on pistons with a ring expander, observing top markings.

Step 17: Stagger ring end gaps at 120 degrees, avoiding the thrust face.

Step 18: Orient piston with arrow toward the timing belt end.

Step 19: Orient rod with bearing oil squirter toward the exhaust side.

Step 20: Install wrist pin and circlips, verifying full seating.

Step 21: Install upper rod bearing shell in the rod big end.

Step 22: Lubricate bearing surface with assembly lube.

Step 23: Install piston in ring compressor and clamp.

Step 24: Lower piston and rod assembly into the cylinder from above.

Step 25: Tap the piston in with the wooden handle of a hammer.

Step 26: Guide the rod onto the crank journal as the piston enters the bore.

Step 27: Install rod cap with lubed bearing in original orientation.

Step 28: Torque ARP2000 rod bolts using stretch gauge to 0.0063 inch.

Step 29: Verify the rod moves freely without binding.

Step 30: Measure rod side clearance with a feeler gauge.

13.6 — Critical Notes

WARNING — Oil squirter alignment must be verified visually before first start. Squirters must aim at the underside of the piston crown, not the cylinder wall.

WARNING — Rod bolt stretch measurement is non-negotiable for ARP2000. Torque-only installations have a documented failure history at high cylinder pressure.

WARNING — Never reuse single-use stretch bolts. Head bolts, main bolts, rod bolts, crank pulley bolt, and flywheel bolts are all single use.

NOTE — Bore taper and out-of-round tolerances from the piston manufacturer are stricter than OEM specification. Always follow the manufacturer specification.

5. Torque Master

Consolidated torque reference for every fastener called out in the technical chapters. Sorted by subsystem.

© 2026 Example Garage

Engine Internal

Fastener	Torque	Method	Reuse
Cylinder head bolts OEM M11	40 Nm + 90 + 90 deg	Stretch	No
Cylinder head studs ARP	80 ft lbs (3 steps)	Equal steps	Yes
Main bearing bolts OEM	65 Nm + 90 deg	Stretch	No
Rod bolts OEM	30 Nm + 90 deg	Stretch	No
Rod bolts ARP2000	Stretch 0.0063 inch	Gauge	Yes
Crank pulley bolt	90 Nm + 90 deg	Stretch	No
Flywheel bolts	60 Nm + 90 deg	Stretch	No
Oil pan to block	15 Nm	Center out	Yes
Cam cap bolts	10 Nm	Sequence	Yes
Cam gear bolt	100 Nm	Single use	No
Spark plugs	30 Nm	Standard	Yes

Turbocharger and Exhaust

Fastener	Torque
Exhaust manifold nut	25 Nm
Turbo to T3 manifold	35 Nm
Wastegate flange	35 Nm
Downpipe V-band	30 Nm
Oil feed banjo	22 Nm
Oil drain flange	10 Nm
O2 sensor	55 Nm

Drivetrain

Fastener	Torque	Reuse
Bell housing to block	80 Nm	Yes
Engine mount to chassis	40 Nm + 90 deg	No
Pendulum mount	60 Nm + 90 deg	No
Pressure plate to flywheel	20 Nm	Yes
Axle hub nut	50 Nm + 45 deg	No (stake)
Wheel lug bolts	120 Nm	Yes

© 2026 Example Garage

6. Index

Demonstrates the appendix index pattern for the workshop manual template.

A

Step 1: **ARP rod bolts** — Group 13
Step 2: **Assembly lube** — Group 13
Step 3: **Axle hub nut torque** — Group 92

B

Step 4: **Bearing clearance** — Group 13
Step 5: **Block preparation** — Group 13
Step 6: **Bore measurement** — Group 13

C

Step 7: **Cam gear bolt** — Group 92
Step 8: **Chapter structure** — Group 02
Step 9: **Compression ratio** — Group 05
Step 10: **Crank endplay** — Group 13
Step 11: **Crankshaft installation** — Group 13

D

Step 12: **Diagonal cell rendering** — Group 13
Step 13: **Dial indicator usage** — Group 13

F

Step 14: **Firing order** — Group 05
Step 15: **Flywheel torque** — Group 92
Step 16: **Foreword** — Group 01
Step 17: **Fluid capacities** — Group 05

H

Step 18: **Head bolt torque** — Group 92
Step 19: **How to use** — Group 02

M

Step 20: **Main bearing torque** — Group 13, 92
Step 21: **Manual structure overview** — Group 01

© 2026 Example Garage

O

Step 22: **Oil pan torque** — Group 92

Step 23: **Oil squirter alignment** — Group 13

P

Step 24: **Piston installation** — Group 13

Step 25: **Plastigauge use** — Group 13

R

Step 26: **Ring end gap** — Group 13

Step 27: **Rod bolt stretch** — Group 13, 92

S

Step 28: **Spark plug torque** — Group 92

Step 29: **Special tools** — Group 13

Step 30: **Stretch bolts** — Group 13, 92

T

Step 31: **Technical data** — Group 05

Step 32: **Tightening torques master** — Group 92

Step 33: **Turbo flange torque** — Group 92

Service Group Cross Reference

Group	Title	Part
1	Foreword	1 General Information
2	How to Use This Manual	1
5	Technical Data Summary	1
13	Crankshaft and Cylinder Block	2 Engine Mechanical
92	Torque Specifications Master	8 Technical Appendices
98	Alphabetical Index	8

© 2026 Example Garage

Reference Tables

Consolidated reference tables extracted from the documentation.

Foreword

Item	Detail
Edition	1.0
Publication date	Example
Format	BriefKit manual template, navy preset
License	CC0 1.0

How To Use

Section	Purpose
Technical Data	Key dimensions, clearances, and operating parameters
Special Tools	Factory tool numbers and aftermarket equivalents
Tightening Torques	Consolidated torque specification table
Removing	Disassembly procedure as numbered steps
Installing	Reassembly procedure as numbered steps
Testing and Checking	Measurements and acceptance criteria
Notes	Technical notes, safety callouts, cross references

Technical Data

Item	Specification
Displacement	2008 cc (built)
Bore	83.01 mm
Stroke	92.80 mm
Compression ratio	9.3:1
Firing order	1-3-4-2
Target peak power	500 wheel horsepower
Target peak torque	520 Nm at wheels
Fuel	E85 flex
Redline	7200 rpm

© 2026 Example Garage

Technical Data

Fluid	Capacity	Specification
Engine oil with filter	4.2 L	5W40 full synthetic
Coolant	7.2 L	OAT pink
Transmission	2.3 L	75W90 GL4
Brake fluid	1.0 L	DOT 4

Technical Data

Parameter	Target
Peak boost pressure	25 psi
Peak EGT under load	< 950 C
Peak AFR on E85	11.8:1
Maximum ignition retard	0 degrees

Engine Block

Item	Specification
Block material	Grey cast iron
Built bore	83.01 mm ± 0.005 mm
Built stroke	92.80 mm
Built displacement	2008 cc
Main bearing radial clearance	0.018 to 0.038 mm
Rod bearing radial clearance	0.015 to 0.040 mm
Crankshaft endplay	0.070 to 0.230 mm
Rod side clearance	0.10 to 0.30 mm
Piston to wall (forged)	0.089 to 0.114 mm
Top ring end gap (turbo)	0.22 to 0.42 mm

Engine Block

Tool	Purpose
Rod bolt stretch gauge	ARP rod bolt installation
Plastigauge set	Bearing clearance check
Outside micrometer 0 to 100 mm	Crank journal measurement
Dial bore gauge	Cylinder measurement
Ring compressor 83 mm	Piston installation
Dial indicator with magnetic base	Crank endplay

© 2026 Example Garage

Engine Block

Fastener	Torque	Method	Reuse
Main bearing bolt OEM stretch	65 Nm + 90 deg	Stretch	No
Rod bolt OEM stretch	30 Nm + 90 deg	Stretch	No
Rod bolt ARP2000	Stretch 0.0063 inch	Gauge	Yes
Crank pulley bolt	90 Nm + 90 deg	Stretch	No
Flywheel to crank	60 Nm + 90 deg	Stretch	No
Oil pump to block	15 Nm	Standard	Yes
Oil pan to block	15 Nm	Center out	Yes

Torque Master

Fastener	Torque	Method	Reuse
Cylinder head bolts OEM M11	40 Nm + 90 + 90 deg	Stretch	No
Cylinder head studs ARP	80 ft lbs (3 steps)	Equal steps	Yes
Main bearing bolts OEM	65 Nm + 90 deg	Stretch	No
Rod bolts OEM	30 Nm + 90 deg	Stretch	No
Rod bolts ARP2000	Stretch 0.0063 inch	Gauge	Yes
Crank pulley bolt	90 Nm + 90 deg	Stretch	No
Flywheel bolts	60 Nm + 90 deg	Stretch	No
Oil pan to block	15 Nm	Center out	Yes
Cam cap bolts	10 Nm	Sequence	Yes
Cam gear bolt	100 Nm	Single use	No
Spark plugs	30 Nm	Standard	Yes

Torque Master

Fastener	Torque
Exhaust manifold nut	25 Nm
Turbo to T3 manifold	35 Nm
Wastegate flange	35 Nm
Downpipe V-band	30 Nm
Oil feed banjo	22 Nm
Oil drain flange	10 Nm
O2 sensor	55 Nm

© 2026 Example Garage

Engineering Variant Sections

Safety Warnings Summary

No structured safety warnings found — always consult source documentation and relevant standards before performing engineering work.

Specifications Quick-Reference

Engineering Specifications

Parameter	Value	Notes
Bore	83.01 mm	
Stroke	92.80 mm	
Target peak torque	520 Nm at wheels	
Redline	7200 rpm	
Peak boost pressure	25 psi	
Built bore	83.01 mm ± 0.005 mm	
Built stroke	92.80 mm	
Main bearing radial clearance	0.018 to 0.038 mm	
Rod bearing radial clearance	0.015 to 0.040 mm	
Crankshaft endplay	0.070 to 0.230 mm	
Rod side clearance	0.10 to 0.30 mm	
Piston to wall (forged)	0.089 to 0.114 mm	
Top ring end gap (turbo)	0.22 to 0.42 mm	
Outside micrometer 0 to 100 mm	Crank journal measurement	
Ring compressor 83 mm	Piston installation	

Standards Reference

No standards citations found — see source documents.

© 2026 Example Garage

Example Garage

"OEM-style workshop manual demonstration"

Example Garage

This document was generated on 14 October 2026.

© 2026 Example Garage.

Dedicated to the public domain under CC0 1.0.

Generated by briefkit